

IMPLEMENTATION_TRACKER

HelixCode Implementation Tracker

CRITICAL IMPLEMENTATION PLAN

Started: 2025-01-08
Status: IN PROGRESS
Priority: CRITICAL FIXES FIRST

IMPLEMENTATION TASKS

WEEK 1: CRITICAL SECURITY & INFRASTRUCTURE FIXES

COMPLETED

- ☐ Security Audit Complete - Identified critical vulnerabilities
- ☐ Implementation Tracker Created - This document

IN PROGRESS

- ☐ SSH Security Hardening (CRITICAL)
- ☐ Distributed System Redesign (CRITICAL)
- ☐ Real-time Data Integration (CRITICAL)

PENDING

- ☐ Connection Pooling Implementation
- ☐ Load Balancing System
- ☐ Consensus Mechanism Implementation

DETAILED PROGRESS LOG

2025-01-08 - Day 1

Morning Session (09:00 - 12:00)

Objective: Fix critical SSH security vulnerabilities

Status: IN PROGRESS

File: internal/worker/ssh_pool.go

Issues Found:

// CRITICAL SECURITY ISSUE - Line 287
ssh.InsecureIgnoreHostKey() // ALLOWS MAN-IN-THE-MIDDLE ATTACKS

Actions Taken: 1. Backed up original file 2. Created security fix implementation plan 3. Started implementing secure host key verification

Next Steps: - Complete secure SSH implementation - Add certificate pinning - Implement worker isolation

Afternoon Session (13:00 - 17:00)

Objective: Distributed system architecture redesign

Status: IN PROGRESS

Files: internal/worker/, internal/task/

Architecture Changes Needed: 1. Consensus mechanism implementation 2. Load balancing with capability matching 3. Worker auto-recovery system 4. Task migration logic

TEST COVERAGE TRACKER

Current Test Bank Analysis

Test Types Supported: - Unit Tests (./tests/unit/) - Integration Tests (./tests/integration/) - E2E Tests (./tests/e2e/) - Performance Tests (./tests/performance/) - Mock Services (./tests/mocks/)

Test Execution Scripts: - ./scripts/run-tests.sh - Master test runner - ./scripts/run-all-tests.sh - Comprehensive suite - docker-compose.test.yml - Test environment

 **Coverage Requirements:** - Target: 95%+ code coverage - Current: 87% (pre-fixes) - Goal: Maintain >95% during implementation

PROGRESS METRICS

Critical Fixes Progress

- **SSH Security:** 10% complete (analysis done)
- **Distributed System:** 5% complete (planning done)
- **Real-time Data:** 0% complete (not started)
- **UI/UX Fixes:** 0% complete (not started)

Test Coverage

- **Before:** 87% coverage
- **Current:** [TBD after fixes]
- **Target:** 95%+ coverage

Files Modified

- **Security:** 0 files modified yet
 - **Infrastructure:** 0 files modified yet
 - **Tests:** 0 new tests yet
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NEXT SESSION PLAN (Tomorrow)

Priority 1: Complete SSH Security

- Implement secure host key verification
- Add worker isolation sandboxing
- Create SSH security tests
- Update documentation

Priority 2: Start Distributed System

- Implement Raft consensus algorithm
- Add worker capability detection
- Create load balancing algorithms
- Design task migration logic

Priority 3: Test Infrastructure

- Create security test suite
 - Add distributed system integration tests
 - Update performance benchmarks
 - Create chaos engineering tests
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NOTES & DECISIONS

Security Decisions

1. **SSH Implementation:** Moving from `InsecureIgnoreHostKey()` to certificate pinning
2. **Worker Isolation:** Implementing sandboxing for each worker
3. **Authentication:** Adding mTLS between components

Architecture Decisions

1. **Consensus:** Using Raft algorithm for leader election
2. **Load Balancing:** Capability-based assignment with weighted round-robin
3. **Failure Recovery:** Automatic task migration with checkpoint restoration

Testing Strategy

1. **Security:** Penetration testing suite
 2. **Distributed:** Chaos engineering with failure injection
 3. **Performance:** Load testing with realistic workloads
 4. **Integration:** Full workflow testing
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RELATED FILES

- **Main Code:** `/Users/milosvasic/Projects/helix_code/helix_code/`
 - **Tests:** `/Users/milosvasic/Projects/helix_code/helix_code/tests/`
 - **Analysis:** `/Users/milosvasic/Projects/helix_code/COMPREHENSIVE_ANALYSIS_REPORT.md`
 - **Tracker:** `/Users/milosvasic/Projects/helix_code/IMPLEMENTATION_TRACKER.md`
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RISKS & BLOCKERS

Current Blockers

- None identified yet

Potential Risks

- SSH security changes might break existing worker connections
- Distributed system changes require extensive testing
- Real-time data integration requires WebSocket stability

Mitigation Strategies

- Backward compatibility during migration
- Comprehensive test coverage
- Staged rollout with monitoring

Last Updated: 2025-01-08 17:00